Experiment No: 2 - Shape and Reshaping of NumPy Arrays

Aim

To study the dimensions, shape, size, reshaping, flattening, and transpose operations of NumPy arrays.

Software Required

Python 3.x, NumPy Library, Jupyter Notebook / Google Colab / Python IDLE

Theory

NumPy provides attributes and functions to determine the dimensions, shape, and size of arrays and to perform reshaping, flattening, and transpose operations efficiently.

a) Dimensions of NumPy Array

Program:

```
import numpy as np
arr = np.array([[1,2,3],[4,5,6]])
print(arr.ndim)
```

Output:

2

b) Shape of NumPy Array

Program:

```
import numpy as np
arr = np.array([[1,2,3],[4,5,6]])
print(arr.shape)
```

Output:

(2, 3)

c) Size of NumPy Array

Program:

```
import numpy as np
arr = np.array([[1,2,3],[4,5,6]])
print(arr.size)
```

Output:

6

d) Reshaping a NumPy Array

Program:

```
import numpy as np
arr = np.array([1,2,3,4,5,6])
new_arr = arr.reshape(2,3)
print(new_arr)
```

Output:

```
[[1 2 3]
 [4 5 6]]
```

e) Flattening a NumPy Array

Program:

```
import numpy as np
arr = np.array([[1,2,3],[4,5,6]])
flat = arr.flatten()
print(flat)
```

Output:

```
[1 2 3 4 5 6]
```

f) Transpose of a NumPy Array

Program:

```
import numpy as np
arr = np.array([[1,2,3],[4,5,6]])
print(arr.T)
```

Output:

```
[[1 4]
 [2 5]
 [3 6]]
```

Result

Thus, the dimensions, shape, size, reshaping, flattening, and transpose operations of NumPy arrays were successfully performed.